Light

GCSE Computer Science - Embedded Systems (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Embedded systems are often described as 'hidden computers'. What does this mean?

- ○ **A.** They are invisible to the human eye
- ○ **B.** They must be physically hidden underground
- ○ **C.** They are built into products where users may not realise a computer is present
- ○ **D.** They are too expensive for consumers to see

Q2 (1 mark)

Embedded systems typically use simpler processors than desktop computers. Why is this?

- ○ **A.** Simpler processors are more powerful
- ○ **B.** Dedicated tasks require less processing capability than running multiple general applications
- ○ **C.** Complex processors do not exist yet
- ○ **D.** Desktop computers do not use processors

Q3 (1 mark)

A microwave oven has a controller that manages cooking time, power levels, and the turntable motor. This controller is:

- ○ **A.** A supercomputer
- ○ **B.** A general-purpose laptop
- ○ **C.** An embedded system
- ○ **D.** A games console

Q4 (1 mark)

A fitness tracker worn on the wrist monitors heart rate, counts steps, and tracks sleep. What type of system controls this device?

- ○ **A.** A mainframe computer
- ○ **B.** An embedded system
- ○ **C.** A database server
- ○ **D.** A spreadsheet application

Q5 (1 mark)

When embedded systems are mass-produced, they typically have lower unit costs than general-purpose computers. Why?

- ○ **A.** Components are selected specifically for the required function, eliminating unnecessary hardware
- ○ **B.** Mass production always increases costs
- ○ **C.** Embedded systems use more expensive components
- ○ **D.** General-purpose computers have no components

Q6 (1 mark)

An ATM (cash machine) dispenses money and processes transactions. The controller managing these operations is:

- ○ **A.** A word processor
- ○ **B.** A social media platform
- ○ **C.** An embedded system
- ○ **D.** A video game

Q7 (1 mark)

Embedded systems typically have interfaces designed specifically for their application rather than standard keyboards and mice. Why?

- ○ **A.** Users interact with the product's function, not with a computer, so interfaces match the application
- ○ **B.** Keyboards and mice are too cheap to use

- ○ **C.** Embedded systems cannot connect to any input device
- ○ **D.** Standard interfaces would make the system more reliable

Q8 (1 mark)

A lift (elevator) controller manages floor selection, door operation, and safety systems. This is an example of:

- ○ **A.** A mobile phone application
- ○ **B.** An embedded system
- ○ **C.** A web browser
- ○ **D.** A music streaming service

Q9 (1 mark)

Software updates for embedded systems are typically less frequent than for desktop computers. Why?

- ○ **A.** Dedicated functions rarely need new features, and thorough testing before release minimises bugs requiring patches
- ○ **B.** Embedded systems have no software at all
- ○ **C.** Updates are released daily for all embedded systems
- ○ **D.** Users constantly demand new embedded software features

Q10 (1 mark)

A digital hearing aid processes sound and amplifies specific frequencies for the wearer. The processor controlling this is:

- ○ **A.** A video projector
- ○ **B.** A desktop computer
- ○ **C.** A cloud server
- ○ **D.** An embedded system

Q11 (1 mark)

Embedded systems often provide deterministic behaviour. What does this mean?

- ○ **A.** Given the same inputs, they always produce the same outputs within predictable time limits
- ○ **B.** They behave differently each time they run
- ○ **C.** They determine their own random behaviour
- ○ **D.** They cannot respond to any inputs

Q12 (1 mark)

A smart speaker responds to voice commands and plays music from streaming services. The device contains:

- ○ **A.** No processing capability whatsoever
- ○ **B.** A general-purpose desktop PC
- ○ **C.** Only a simple speaker with no electronics
- ○ **D.** An embedded system for voice processing and audio output

Q13 (2 marks)

A petrol pump at a filling station dispenses fuel and calculates the cost. Describe two functions the embedded system performs.

- ○ **A.** Playing video games and browsing social media
- ○ **B.** Editing photographs and creating documents
- ○ **C.** Measuring fuel flow precisely and calculating total cost based on current fuel prices
- ○ **D.** Streaming movies and recording television

Q14 (2 marks)

Embedded systems typically boot instantly rather than taking minutes like desktop computers.
Explain why this is possible and why it matters.

- ○ **A.** Instant boot is impossible for any computer system
- ○ **B.** Boot time has no importance for any application
- ○ **C.** Desktop computers boot faster than embedded systems
- ○ **D.** Simple dedicated software stored in ROM/flash loads immediately without complex initialisation; this matters because many applications cannot wait for lengthy startup

Q15 (2 marks)

A blood glucose meter allows diabetic patients to test their blood sugar levels.
Explain why embedded system characteristics suit this medical device.

- ○ **A.** Portability requires small size and battery power; reliability is essential for medical decisions; simple operation allows use by patients without technical training
- ○ **B.** Blood glucose meters need to run spreadsheet software
- ○ **C.** These devices require keyboard and mouse input
- ○ **D.** Blood glucose meters must be connected to the internet at all times

Q16 (2 marks)

Railway signal systems control train movements across networks. Identify two requirements that make embedded systems essential.

- ○ **A.** Signals need to display advertisements and play music
- ○ **B.** Railway systems only work when manually operated
- ○ **C.** Signals must run word processing software
- ○ **D.** Absolute reliability to prevent collisions and real-time response to track occupancy sensors

Q17 (2 marks)

Many embedded systems use microcontrollers rather than separate processor, memory, and I/O chips.
Explain the advantages of this integration.

- ○ **A.** Separate components are always better than integrated ones
- ○ **B.** Integration makes devices larger and more expensive
- ○ **C.** Microcontrollers combine processor, memory, and I/O on one chip, reducing size, cost, power consumption, and simplifying circuit

board design

- ○ **D.** Microcontrollers cannot perform any useful processing

Q18 (2 marks)

A digital camera captures photographs and stores them on memory cards.
Describe two functions its embedded system performs.

- ○ **A.** Making phone calls and sending text messages
- ○ **B.** Processing sensor data into image files and controlling autofocus by analysing scene sharpness
- ○ **C.** Running office productivity software
- ○ **D.** Playing video games online

Q19 (2 marks)

Many embedded systems include a 'watchdog timer' that resets the system if software stops responding.
Explain why this feature is important.

- ○ **A.** Watchdog timers are only used for telling time
- ○ **B.** Watchdog timers slow down system performance
- ○ **C.** If software crashes or enters an infinite loop, the watchdog automatically restarts the system, restoring normal operation without human intervention
- ○ **D.** This feature is never used in any embedded system

Q20 (2 marks)

A car's airbag controller detects collisions and deploys airbags to protect occupants.
Why must this embedded system have extremely fast response times?

- ○ **A.** Response time does not matter for safety systems
- ○ **B.** Collisions occur in milliseconds - airbags must deploy before occupants hit the steering wheel or dashboard, requiring detection and deployment within 15-30 milliseconds
- ○ **C.** Airbags work best when deployed slowly over several minutes
- ○ **D.** Fast response would make the airbag less effective

Q21 (2 marks)

Embedded software is often developed on desktop computers but runs on different processors.
Explain how this is achieved.

- ○ **A.** Desktop software automatically works on all processors without changes
- ○ **B.** Embedded systems can only run software written directly in machine code by hand
- ○ **C.** Cross-compilers translate source code into machine code for the target embedded processor, allowing development on powerful desktop computers
- ○ **D.** Development computers and target systems must always use identical processors

Q22 (2 marks)

A smart electricity meter measures energy consumption and communicates readings to the supplier.
Identify two embedded system characteristics this requires.

- ○ **A.** Ability to run video games and edit photographs
- ○ **B.** Keyboard and mouse for user interaction
- ○ **C.** High-resolution display for watching films
- ○ **D.** Accurate continuous measurement over many years and reliable communication to transmit usage data

Q23 (2 marks)

Debugging embedded systems can be more challenging than debugging desktop software.
Explain why this is the case.

- ○ **A.** Desktop software is impossible to debug
- ○ **B.** Embedded systems have more debugging tools than desktop computers
- ○ **C.** Limited display and interface options make it hard to see what the program is doing; debugging tools may alter timing behaviour, masking real-time bugs
- ○ **D.** Debugging is never necessary for embedded software

Q24 (3 marks)

An electric vehicle charging station delivers power to cars and processes payments.
Analyse the embedded system requirements for safe and reliable operation.

- ○ **A.** The system must monitor charging current and temperature for safety, communicate with vehicle battery management systems to deliver appropriate power, and securely process payment transactions
- ○ **B.** Charging stations require no electronic control systems
- ○ **C.** Charging stations only need to supply raw electricity with no control
- ○ **D.** Payment processing is unnecessary for charging stations

Q25 (3 marks)

A railway ticket machine sells tickets using touchscreen, card payment, and printed receipts.
Analyse the embedded system challenges for public deployment.

- ○ **A.** Ticket machines face no challenges in public deployment
- ○ **B.** Ticket machines require constant human supervision to operate
- ○ **C.** Public deployment makes embedded systems simpler to design
- ○ **D.** The system must withstand vandalism and weather exposure, recover automatically from software errors without staff intervention, and process payments securely to prevent fraud

Q26 (3 marks)

Embedded systems can respond to events using interrupts or polling. Analyse the differences and when each approach is appropriate.

- ○ **A.** There is only one way for embedded systems to respond to events
- ○ **B.** Interrupts immediately signal the processor when events occur, enabling fast response with low power; polling repeatedly checks for

events, using more power but being simpler to implement - choice depends on response time and power requirements

○ **C.** Polling always provides faster response than interrupts

○ **D.** Neither interrupts nor polling are used in embedded systems

Q27 (3 marks)

A wind turbine controller manages blade pitch, rotation speed, and power generation.
Analyse the embedded system requirements for this industrial application.

○ **A.** Wind turbines require no electronic control systems

○ **B.** Maintenance can easily be performed on turbines at any time

○ **C.** Wind turbines only need simple on/off switches

○ **D.** The system must operate reliably for decades with minimal maintenance, respond to changing wind conditions in real-time, and implement safety shutdowns for extreme weather while maximising power generation

Q28 (3 marks)

Safety-critical embedded systems in aircraft require extensive certification.
Analyse why this certification process exists and what it involves.

○ **A.** Failures could cause loss of life; certification requires documented development processes, extensive testing at multiple levels, independent verification, and evidence that all potential failure modes have been identified and mitigated

○ **B.** Aircraft embedded systems require no special testing or approval

○ **C.** Certification is a quick and simple process that takes a few days

○ **D.** Only the final product needs testing, not the development process

Q29 (3 marks)

A 3D printer builds objects by depositing material layer by layer under computer control.
Analyse the embedded system requirements for precise fabrication.

○ **A.** 3D printers require no computational control

○ **B.** The system must precisely control stepper motors for sub-millimetre positioning, coordinate multiple axes simultaneously, manage extruder temperature precisely, and interpret complex geometry data from design files

○ **C.** Only simple on/off control is needed for 3D printing

○ **D.** Manual control produces better results than embedded systems

Q30 (3 marks)

Embedded systems often remain in service long after the components used to build them become obsolete.
Analyse the challenges this creates and how they are addressed.

○ **A.** Replacement components may become unavailable; solutions include stockpiling spare parts, designing for component substitution, and planning system upgrades before critical components are discontinued

○ **B.** Component obsolescence never affects embedded systems

○ **C.** Embedded systems should be replaced annually regardless of function

○ **D.** All electronic components remain available indefinitely

Total: **55** marks • Time: **30 min**

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